

Do Fusion Operators Matter, and Does the Answer Survive Preprocessing? A Parameter-Matched Multiverse Comparison of Multimodal CNN Fusion for EEG Seizure Detection

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Abstract

Convolutional neural networks for EEG seizure detection commonly combine two representations of the same signal, the raw waveform and its spectrogram, using a fusion operator such as early, late, gated, attention, or score level fusion. Published work routinely reports that a particular operator outperforms the alternatives, but the compared models typically differ in more than the operator: parameter count, training protocol, and evaluation design vary alongside it, and the paired statistical tests used to support such claims commonly assume independent measurements that repeated cross validation violates. We hold backbone architecture, training protocol, and parameter budget fixed to within 1.3 percent and vary only the fusion operator, then evaluate the resulting differences with a resampling corrected paired test, Holm correction, and two one sided equivalence testing, on two corpora: the Bonn EEG dataset (5x5 repeated stratified cross validation) and CHB-MIT (leave one subject out over 23 individuals). On Bonn, no pair among late, gated, attention, and score level fusion separates under the corrected test in any of three task formulations, and the pairs are equivalent within an F1 margin of 0.02 to 0.03; early fusion is the one operator that departs, and only downward. On CHB-MIT, none of the same five operators separate either, but between subject variability is large enough that equivalence itself cannot be established with confidence, a materially different kind of negative result that we report separately from Bonn’s. Two further findings bear on evaluation practice generally. First, a simulation shows the naive paired Wilcoxon test commonly used in this literature has a 38 percent false positive rate under repeated cross validation; correcting for it collapses most of the differences we originally observed. Second, CPU thread count alone, a setting with no bearing on the scientific question, shifts per fold macro F1 by up to 0.09 on Bonn and 0.21 on CHB-MIT, a magnitude comparable to or larger than the differences between the operators under comparison; re-running CHB-MIT after correcting the subject grouping and unifying the optimiser left every conclusion intact but reordered the eleven models almost arbitrarily (Spearman $\rho = 0.45$ between the two runs). We also report and correct three errors we found along the way: a training bug that let some models collapse to a trivial single class solution while reporting high AUC, a labelling error in CHB-MIT’s own seizure annotations that a sequence naive parser would turn into a nonexistent 6316 second seizure, and a subject grouping error in our own CHB-MIT evaluation, where two recording cases from the same person were treated as different subjects. We argue that the absence of a detectable difference between fusion operators in this literature is not yet established as a real finding so much as it is obscured by evaluation practices too imprecise to resolve differences at this scale, and that equivalence testing, corrected resampling statistics, and reporting of runtime determinism should be standard rather than exceptional in this line of work.

1. Introduction

Electroencephalography (EEG) records the brain’s electrical activity over time. During an epileptic seizure this signal changes characteristically: amplitude rises, the waveform becomes

more rhythmic, and power concentrates in particular frequency bands. Automatic seizure detection is therefore a signal classification problem, and convolutional neural networks (CNNs) are the dominant modelling approach to it.

A recurring design pattern in this literature represents the same EEG segment two ways at once: the raw waveform, processed by a 1D CNN, and its spectrogram, processed by a 2D CNN. The two branches are combined by a fusion operator, most commonly one of early fusion (combining inputs or early features), late fusion (combining final feature vectors before classification), gated fusion (a learned per branch weight), attention based fusion, or score level fusion (averaging the two branches' independent predictions). Published comparisons of these operators typically report that the authors' chosen operator wins. The comparisons, however, usually differ in more than the operator under study: model capacity, training schedule, and data handling vary alongside it. When a larger fusion model outperforms a smaller single branch model, it is not clear whether the gain comes from the fusion mechanism or simply from the added capacity.

This paper asks a narrower and more mechanical question than most work in this space: when architecture size, training protocol, and data are held fixed, and only the fusion operator is varied, does the operator choice produce a difference that survives a statistical test appropriate to the way the data was resampled? We treat this as a comparative and methodological study rather than a proposal for a new architecture, and we hold ourselves to the same scrutiny we apply to the literature: every reported number in this manuscript traces to a file under `results_v2/`, and every claim about what has or has not been shown before is qualified by how the underlying literature search was actually conducted (Section 2 and `docs/09`, `docs/14`).

1.1 Contributions

1. A parameter matched benchmark of five fusion operators plus single branch and classical feature controls, on the Bonn EEG corpus, under three task formulations, evaluated with 5x5 repeated stratified cross validation (Section 4).
2. An audit of the statistical practice common in this literature: a simulation quantifying the false positive rate of the naive paired test under repeated cross validation, and a corrected testing framework (resampled t test, Holm correction, equivalence testing, and a correlated Bayesian t test) applied throughout (Section 3.5).
3. External validation on CHB-MIT with subject wise leave one subject out evaluation, which the Bonn corpus does not support because it publishes no subject identifiers (Section 5), with subjects grouped by person rather than by recording case, after finding that two of CHB-MIT's 24 cases are documented as the same individual (Section 5.2).
4. Two methodological findings that generalise beyond this specific comparison: naive paired testing under repeated cross validation and CPU thread count as an unreported source of variance comparable in magnitude to the effect under study (Section 6).
5. A corrected, order based parser for CHB-MIT's seizure annotation files, after finding that the standard number based parsing is vulnerable to a labelling error present in the released files (Section 5.1).
6. A preprocessing and training multiverse (Section 3.6): nine preprocessing pipelines, including three artefact removal algorithms, each crossed with three seed sets, to ask whether conclusions about fusion operators, not only their absolute performance, are robust to preprocessing decisions and to stochastic training (results pending, Section 5.6).

2. Related Work

Table 1 summarises the closest published work to this comparison, restricted to what we have verified through Crossref by DOI. It draws on a systematic search run on 3 October

2026 under a protocol fixed beforehand (docs/14_SISTEMATIK_TARAMA_PROTOKOLU.md): Scopus, Web of Science, IEEE Xplore and arXiv returned 902 records, 555 after deduplication, of which 128 passed title and abstract screening; the studies closest to ours were read in full, and every claim Table 1 makes about them is traced to a section or table of the paper (paper/literature/extraction.csv, docs/15_LITERATUR_SONUCLARI.md). Title and abstract screening was done by an AI assistant applying the protocol’s criteria; an independent AI pass over a random 20 percent agreed on 91.9 percent of records (Cohen’s kappa 0.81). A human second screening and 122 full texts are still outstanding (Section 7).

Table 1: Closest related work.

Work	Contribution	What it leaves open
Andrzejak et al. (2001)	Introduces the Bonn EEG corpus	No per segment subject identifiers
Acharya et al. (2018)	Early deep CNN for Bonn seizure detection	Single architecture, no operator comparison
Truong et al. (2018)	CNN over spectrograms, scalp and intracranial	No raw signal branch to fuse against
Wang et al. (2020)	Compares 1D and 2D CNNs	Compared separately, not fused
Chatzichristos et al. (2020)	Attention gated multi view fusion	One operator, no alternatives compared
Das et al. (2024)	1D+2D fusion of decomposed EEG	Fixed design, no operator ablation
Huang et al. (2024)	Multi head attention fusion	Reports gains without matching parameter count
Golrizkhatami and Acan (2018)	Three level fusion for ECG	Different signal domain, related fusion methodology
Narotamo et al. (2024)	Compares 1D and 2D representations plus early, late, and joint fusion, the same design question as ours	ECG, not EEG; no parameter matching or corrected/equivalence statistics
Roy et al. (2019), Shoeibi et al. (2021), Xu et al. (2024)	Reviews of deep learning for EEG	Document inconsistent evaluation protocols, do not resolve them
Ali et al. (2024)	Shows CHB-MIT results depend heavily on evaluation choices	Concerns evaluation of CHB-MIT generally, not fusion operators
Dan et al. (2024), SzCORE	Standard datasets, event based scoring and metrics for EEG seizure detection	An evaluation framework, not a fusion operator comparison
Rheude et al. (2025)	Multimodal complexity often does not pay off	General multimodal setting, not EEG
Daşdemir and Örnek (2025)	Self described fair, protocol controlled comparison of two fusion strategies for EEG	Seizure <i>prediction</i> , not detection; two operators; reports a 0.20 point accuracy difference (97.50 vs 97.70) without a stated statistical test or parameter matching
An et al. (2026), fastSeizureNet	Compares five fusion strategies for EEG seizure detection , within and across patients, with several backbones (semi-supervised)	Fuses hand-crafted feature knowledge with deep features rather than a raw waveform with its spectrogram; the strategies are not parameter matched and are not compared with each other by a statistical test (a Wilcoxon test with Holm correction is used only against prior methods)

Work	Contribution	What it leaves open
Einizade et al. (2023), fAttNet	Compares four fusion operators (majority vote, averaging, concatenation, attention) over raw EEG, wavelet packet and hand-engineered views, leave one subject out on TUH	No parameter matching and no statistical test; the operators lie within one standard deviation of each other
Basheer and Mishra (2026)	Compares concatenation, element-wise addition and KAN gating for seizure detection, with paired t-tests over six seeds	Both streams come from the same raw EEG rather than two representations; uncorrected tests against concatenation only; no parameter matching
Wang et al. (2026), <i>Neurocomputing</i>	Cross-attention fusion of time domain and S-transform views, CHB-MIT and Siena	One ablation against concatenation, patient specific evaluation, no statistical test
Camastra et al. (2026), <i>Brain Sciences</i>	Controlled fusion benchmark, concludes strategy matters more than architecture	Tabular neuroimaging features, not EEG time series; no paired or equivalence testing
Mohamady et al. (2026)	Seven fusion techniques compared on one benchmark	Human activity recognition, not EEG; states this kind of head to head comparison did not previously exist in that field either
Kontras et al. (2026), <i>NeuroAtlas</i>	Reports Bonn is saturated (11 models at AUROC ≥ 0.99) and carries no subject identifiers	Foundation model benchmark, not a fusion operator comparison

Reading this table, fusion operators for EEG seizure detection are proposed frequently, often one at a time against baselines that differ in capacity from the proposed model, and several studies now compare two to five operators (Einizade et al. 2023; An et al. 2026; Basheer and Mishra 2026), but none we read in full matches parameter counts or tests the operators against each other with corrected or equivalence tests. Two studies come closest. Daşdemir and Örnek (2025) holds two branches fixed and compares two fusion points, but on a prediction rather than detection task, without a stated parameter matching procedure or significance test on the 0.20 point difference it reports. An et al. (2026) compare five fusion strategies for seizure detection, the closest design to ours in breadth, but between hand-crafted and deep features rather than between a raw waveform and its spectrogram, without matching the strategies’ parameter counts and without a statistical test between them. We have not found a study that compares three or more fusion operators for EEG seizure detection under a matched parameter budget with paired statistical testing and equivalence testing. This is the gap this paper addresses, stated at this narrower scope rather than as a categorical absence, and subject to revision once the remaining full texts of the systematic search (docs/15) have been read. An independent review across the wider EEG-based multimodal human-computer interface literature is consistent with this being a real gap rather than a search artifact: Lee et al. (2025), surveying over 120 hybrid EEG studies, report that only two explored more than one fusion strategy and only one directly investigated which fusion approach was best, and name fusion strategy optimisation as a direction the field has largely left open.

Two further points from Table 1 shape our design directly. First, NeuroAtlas reports that Bonn’s conventional binary split is saturated, eleven foundation models reaching AUROC at or above 0.99, and recommends treating it as a sanity check rather than a discriminating benchmark; this is not our own inference but an independent finding we build on (Section 4.1, Section 6.3). Second, the same benchmark confirms Bonn publishes no per segment subject identifier, which is why external validation with genuine subject wise cross validation (Section 5) is necessary rather than optional for any claim this paper makes about generalisation across patients.

3. Method

3.1 Architecture

Two branch CNN: a 1D convolutional encoder (channel widths 16, 32, 64) over the raw waveform, and a 2D convolutional encoder of matching depth over the log magnitude short time Fourier transform (STFT) spectrogram of the same segment. The two encoders’ pooled features (dimension 128 each) are combined by one of five interchangeable fusion heads:

- **Early fusion.** The raw signal is passed through a small strided 1D encoder, broadcast across the spectrogram’s time axis, and concatenated as an additional channel before the 2D backbone. We note for precision that this is what the literature calls “early” fusion but is, architecturally, an intermediate fusion: a genuinely input level combination of a 1D waveform and a 2D spectrogram is not well defined, since the two representations do not share a common tensor shape.
- **Late fusion.** The two branches’ pooled feature vectors are concatenated and passed through a small MLP classifier head.
- **Gated fusion.** A learned scalar gate per branch reweights the concatenated features before classification.
- **Attention fusion.** A learned attention mechanism computes per branch weights from both feature vectors jointly.
- **Score level fusion.** The two branches are trained as independent unimodal classifiers; their softmax outputs are blended with a weight selected by grid search on the validation split only, never on test data (`src/train.py: best_blend_weight`).

Reference models: **raw1d** and **spec2d**, the same two backbones trained and classified independently at their default width; **raw1d_wide** and **spec2d_wide**, the same single branch models widened to match the fusion models’ parameter count, which is the correct comparison for asking whether fusion outperforms a unimodal model rather than simply outperforming a smaller one; **logvar**, a one feature logistic regression on the segment’s log variance; and **shallow**, a logistic regression on seven hand crafted features (log variance, line length, and relative power in five frequency bands). The two classical baselines are not deep models and are included to establish how much of any reported performance is attributable to representation learning at all, rather than to the discriminability of the task itself (`src/baselines.py`, `src/chbmit_baselines.py`).

3.2 Parameter matching

Every jointly trained variant (early, late, gated, attention, and the two widened single branch controls) is sized to a common budget with a fusion head hidden dimension solved numerically per operator (`src/models.py: _solve_hidden`) to land within 1.3 percent of the target. On Bonn (single channel input) this budget is approximately 98,500 parameters (late 98,571; gated 98,698; attention 98,544; early 97,507; spec2d_wide 97,775; raw1d_wide 99,795). Score level fusion, being an ensemble of two independently sized unimodal models rather than a jointly parameterised head, is reported at its own natural parameter count (59,302 on Bonn) rather than forced into the same budget, since forcing it there would require shrinking its two constituent classifiers below their own useful capacity. On CHB-MIT, the same budget solved for 18 input channels lands between 102,236 and 103,712 parameters across the same model set (Section 5.2), with a spread of 0.36 percent across the four jointly trained fusion operators (late 102,768; gated 102,921; attention 102,767; early 102,547).

3.3 Preprocessing

Bonn’s segments are band limited at acquisition to 0.53 to 40 Hz (Andrzejak et al. 2001); the spectrogram is computed with a 256 point STFT (window 256, hop 128) and capped at the same 40 Hz. CHB-MIT is not band limited, and we measured (Section 5.1) the largest ictal to interictal power ratio in the gamma band (30 to 70 Hz); the frequency ceiling is therefore raised to 64 Hz for CHB-MIT rather than reused at 40 Hz (`src/chbmit_prep.py`), and line noise removal at the US mains frequency (60 Hz) is implemented as an optional, separately reported arm rather than baked into the default pipeline, since it falls inside the informative gamma band.

Amplitude normalisation on Bonn is treated as an experimental factor with four arms (N0 through N3, `src/config.py: NORM_MODES`) rather than a single fixed choice, because an audit of the original implementation this project began from found the cached spectrogram had in fact been computed from the unnormalised signal while the 1D branch saw a z scored signal, an inconsistency that biased comparisons involving early fusion specifically (Section 6.4). The default arm used throughout (N2) z scores both branches from the same normalised signal.

3.4 Training and evaluation protocol

AdamW optimiser (decoupled weight decay; Loshchilov and Hutter 2019), learning rate 1e-3, weight decay 1e-4, batch size 32, class weighted cross entropy loss, on both corpora. An earlier CHB-MIT run, which we compare against in Section 5.4, used Adam with coupled (L2) weight decay instead. A minimum epoch budget of 20 is enforced before early stopping on validation macro F1 (patience 12, ceiling 60 epochs); this guard was added after an audit found 6 of 55 early runs had frozen in a trivial single class solution while still reporting AUC near 0.96, because early stopping triggered while validation macro F1 was flat rather than genuinely converged (Section 6.4). Seeds are deterministic per (repeat, fold, model) via a hash of those three values (`src/config.py: fold_seed`), removing a dependency on execution order that the original implementation had.

On Bonn, evaluation is 5x5 repeated stratified cross validation (25 paired measurements per model) with an inner 20 percent validation split carved from the training fold, never touching the test fold. On CHB-MIT, evaluation is 23 fold leave one subject out over 24 recording cases, with the same inner validation split constructed subject wise (Section 5.2), since Bonn’s lack of subject identifiers makes this comparison impossible there. All CHB-MIT models were trained on one machine at a fixed thread count of 16 (Section 6.2).

3.5 Statistical framework

Repeated cross validation measurements are not independent: the folds share training data across repeats, which the standard paired test (independent samples, or naive paired Wilcoxon or t test) does not model. We quantified the consequence by simulation (`src/sim_cv_correlation.py`): two statistically identical logistic regression classifiers, differing only in which of two equally informative feature sets they see, evaluated by 5x5 repeated cross validation over 1000 simulated datasets. Naive Wilcoxon rejected the (true) null of no difference in 37.6 percent of simulations, and a naive paired t test in 38.6 percent, against a nominal 5 percent. A resampling corrected t test (following Nadeau and Bengio 2003 and Bouckaert and Frank 2004, which inflate the standard error by the fraction of data shared between the resampled training sets) rejected in 2.3 percent of simulations, close to nominal. This matches an independent, external result: Jafrasteh et al. (2025), studying neuroimaging classifiers rather than EEG fusion specifically, show by a different route that whether a cross validated comparison comes out significant depends on

the number of folds and repeats chosen, which is exactly the dependence structure that makes p-hacking possible under the naive test.

All significance claims in this paper therefore use the corrected test (`src/stats.py:corrected_ttest`), with Holm correction across all pairwise comparisons within a task and metric. Where a null result is claimed, we support it with a two one sided equivalence test computed with the same corrected variance (`corrected_equivalence_bound`, reporting the minimum margin $\delta_{\min}$ at which equivalence would be established at $\alpha = 0.05$), a correlated Bayesian t test (following Corani and Benavoli 2015; Corani et al. 2017) reporting the posterior probability of practical equivalence within a stated region of practical equivalence (ROPE), and a sign test as a distribution free check. We report primary results on macro F1 and Brier score; log loss is reported but treated as secondary, because it proved heavy tailed and unstable across both corpora (Section 4.3, Section 5.2), inflating variance for any model with even a few confidently wrong predictions and correspondingly losing statistical power.

3.6 Preprocessing and training multiverse

The re-run of Section 5.4 showed that settings unrelated to the fusion operator can reorder the operators while leaving the null result intact. We therefore ask a broader question than “which operator is best”: how robust are conclusions about fusion operators to preprocessing decisions and to stochastic training? The design follows the multiverse approach (Steege et al. 2016): a fixed set of defensible analysis choices is crossed and every combination is run, instead of reporting one.

Pipelines. Each pipeline changes one hypothesis driven aspect of preprocessing; windows, labels, subsampling, spectrogram settings (64 Hz ceiling, so input shapes and parameter counts do not change), models, training and evaluation stay fixed (`src/preprocess.py`).

Pipeline	Band-pass	Notch	Artefact handling	Normalisation	Question
P0	none	none	none	z-score	original benchmark (Section 5.3)
P1	0.5 to 40 Hz	none	none	z-score	standard EEG band
P2	0.5 to 70 Hz	60 Hz	none	z-score	information above 40 Hz
P3	1 to 40 Hz	none	none	z-score	low frequency content and drift
P4	0.5 to 40 Hz	none	technical artefact rejection	z-score	artefact rejection
P5	0.5 to 40 Hz	none	none	median and IQR	normalisation
P6a	0.5 to 40 Hz	none	Extended Infomax ICA	z-score	aggressive cleaning
P6b	0.5 to 40 Hz	none	GEDAI	z-score	aggressive cleaning
P6c	0.5 to 40 Hz	none	AMICA	z-score	aggressive cleaning

Filters are zero phase Butterworth filters (order 4, applied forward and backward) run on each continuous recording before it is cut into windows, since filtering 10 s windows separately would place filter transients at every window edge. Every pipeline’s corpus holds exactly the windows of P0, which is checked when it is built, so all pipelines are paired fold by fold.

P4: rejecting technical artefacts, not amplitude. We measured amplitude distributions before choosing a criterion. The median worst channel peak to peak amplitude is 482 μV in non-ictal and 890 μV in ictal windows, so every amplitude threshold removes seizures selectively: 500 μV rejects 72 percent of ictal against 48 percent of non-ictal windows, 1000 μV 44 against

12, and a per subject robust z score above 10 still 21 against 4.5. Amplifier clipping could not serve as a criterion either, because the stored integer data exceed the declared physical range and no recording piles up at a limit. P4 therefore rejects only technical artefacts: a flat channel (standard deviation below $1\ \mu\text{V}$ over the window) or a constant segment of at least 0.5 s on any channel (a signal dropout, at $0\ \mu\text{V}$ in CHB-MIT), detected on the unfiltered signal. This rejects 3.3 percent of ictal and 0.8 percent of non-ictal windows. The difference is not physiological: the 0.5 s before a dropout have the same amplitude in both classes (median about $40\ \mu\text{V}$); dropouts occur in a few recordings that also contain seizures. Rejected windows are removed from training and validation only, so the test set is identical across pipelines.

P6: three artefact removal algorithms. To separate the effect of the algorithm from the rest of the pipeline, P6 is run with Extended Infomax ICA (Lee et al. 1999, as implemented in MNE-Python, Gramfort et al. 2013), GEDAI (Ros et al. 2025), and AMICA (Palmer et al. 2008; Python implementation `jamica`), all on the same 0.5 to 40 Hz signal and per recording. Infomax and AMICA share every setting and the same automatic component rule: 16 components, fitted on every 8th sample, and MNE’s correlation based ocular component detection (`find_bads_eog`, $|z| > 3$) with the four frontal FP channels as EOG proxies, since CHB-MIT has no EOG channel. Sixteen, not eighteen, components are used because the 18 channel bipolar montage contains two closed electrode loops on each side (FP1 to O1 along the temporal and along the parasagittal chain), so the data have rank 16; the remaining two singular values are quantisation noise. Muscle components are not removed, because the gamma band carries the strongest ictal signal in this corpus (Section 3.3). GEDAI is not an ICA: it removes what departs from a leadfield based reference covariance, chosen by its own criterion, so the shared component rule cannot apply to it. Its bundled reference covariance C , defined for a referential 10-05 layout, is mapped onto the bipolar montage as DCD^T , with D the electrode difference matrix (a bipolar difference cancels the common reference), and the channel mean is removed before and restored after cleaning, because GEDAI would otherwise average reference the bipolar channels. GEDAI’s threshold preset was chosen before any classification run and on signal preservation only, as the advisor required: on six subjects its default removed more than half of the power of 39 percent of recordings and of 33 percent of ictal windows, its conservative preset (“auto-”) of 17 and 13 percent, and neither removed more from ictal than from non-ictal windows within a recording (`src/gedai_qc.py`). P6b uses the conservative preset; the default is kept for a robustness report. All three methods were bit for bit reproducible with a fixed seed. Across all 670 recordings with selected windows, the median recording keeps 78 percent of its power under Infomax and AMICA (about 1.5 components removed per recording) and 99.9 percent under GEDAI, but a minority is cleaned heavily by every method: 22, 23 and 21 percent of recordings lose more than half of their power under Infomax, AMICA and GEDAI (43 percent under GEDAI’s default). Compute per one hour recording differs by an order of magnitude (median 18 s for GEDAI, 33 s for Infomax, 213 s for AMICA; 4.7, 8.5 and 73 CPU hours in total); per recording logs are in `results_v2/qc/p6_recording_logs.csv`.

Seeds and device. Every pipeline is trained with three seed sets (`--repeat`), seed set 0 being the one used throughout Section 5. If the ranking of the fusion operators proves unstable across seeds, the affected pipelines are extended to five. All runs of one analysis use one machine, one thread count and one device. Before GPU training is used, its agreement with the CPU is tested on one fold: two GPU runs must repeat, and the CPU to GPU difference at identical seeds must not exceed the difference between two CPU seed sets (`docs/PIPELINE.md`).

Analyses (`src/multiverse.py`, validated on synthetic runs with known effects): (1) performance per pipeline, seed and model, with AUPRC added to the metrics of Section 3.5; (2) rank stability of the five fusion operators, comparing Kendall’s tau between pipelines with Kendall’s tau between seeds of the same pipeline, with a permutation test that permutes pipeline labels within each seed set, and the same comparison among P6a, P6b and P6c to ask whether the

artefact algorithm alone changes the ranking; (3) the Fusion x Preprocessing interaction in a repeated measures ANOVA over folds (Greenhouse-Geisser corrected); (4) variance components for subject, pipeline, fusion, their interaction and seed; (5) robust equivalence, a fusion pair being equivalent in every pipeline under the corrected test of Section 3.5; (6) robustness against performance, the mean macro F1 of each operator against the spread of its pipeline means; and (7) prediction stability, the agreement of window level predictions between pipelines compared with the agreement between seeds.

4. Bonn Results

4.1 Task formulations

The Bonn corpus (Andrzejak et al. 2001) comprises five sets of 100 single channel segments each (23.6 s, 4097 samples, 173.61 Hz): A and B are surface (scalp) recordings from healthy volunteers, C, D, and E are intracranial depth electrode recordings from epilepsy patients, with E capturing ictal activity. We evaluate three label groupings: **T1**, a three class task (Normal = A,B; Interictal = C,D; Ictal = E); **T2**, an intracranial only binary task (C,D vs E) that removes the scalp versus intracranial recording type as a confound; and **T3**, the conventional binary split used in most published work (A,B,C,D vs E). We retain T3 as a reference point rather than a headline result, because a single feature logistic regression (`logvar`) reaches 0.954 macro F1 on it against 0.986 for the best of eleven models (Table 2), independently consistent with NeuroAtlas’s report that this split saturates.

4.2 Main results

Table 2: Bonn, macro F1 and AUC, mean over 25 paired measurements (5x5 repeated stratified CV). Source: `results_v2/T{1,2,3}_*_N2_z_zspec_*/perfold.csv`.

Model	T1 F1	T1 AUC	T2 F1	T2 AUC	T3 F1	T3 AUC	Params
late	0.974	0.999	0.987	1.000	0.986	1.000	98,571
gated	0.973	0.999	0.987	0.999	0.980	1.000	98,698
attention	0.969	0.997	0.985	0.999	0.977	0.998	98,544
spec2d_wide	0.966	0.998	0.984	0.999	0.982	1.000	97,775
score	0.964	0.998	0.981	0.999	0.976	0.999	59,302
raw1d_wide	0.963	0.997	0.973	0.998	0.976	0.992	99,795
spec2d	0.960	0.998	0.979	0.999	0.972	0.999	32,227
shallow	0.954	0.996	0.964	0.995	0.964	0.995	8
raw1d	0.951	0.996	0.964	0.997	0.969	0.995	27,075
early	0.896	0.977	0.923	0.981	0.899	0.965	97,507
logvar	0.646	0.755	0.933	0.986	0.954	0.992	1

Two patterns hold across all three tasks. First, **early fusion is the one operator that separates from the rest, and only downward** (bold in Table 2). Second, on T2 and T3, **naive Wilcoxon finds 20 and 13 “significant” pairs respectively out of 55, and the corrected test finds zero on both** (`results_v2/phase0/summary.csv`); on T1, naive finds 22 and corrected finds 10, all ten involving `logvar`, none involving a pairwise comparison among the deep models (Section 4.3). We treat this collapse from naive to corrected significance as itself a central finding, not a footnote (Section 6.1).

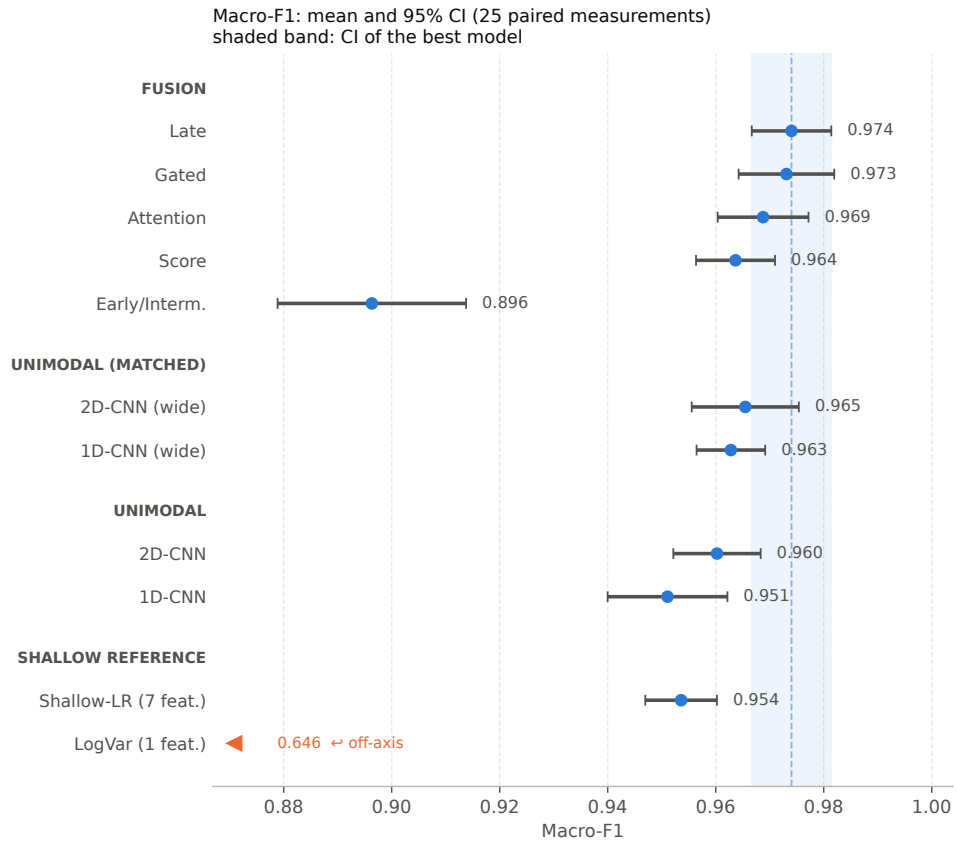


Figure 1: Bonn T1, mean macro F1 with 95% CI per model, grouped by family. Early/intermediate fusion (bold, Table 2) is the one operator visibly separated from the fusion cluster; logvar sits far off axis and is shown pinned to the left edge with its value labelled, rather than compressing the rest of the plot.

Pairwise comparison: macro F1 (Bonn T1) (corrected test)

$\neq$ significant difference (corrected+Holm, $p<0.05$) $\equiv$ equivalent ($\delta_{\min} \leq 0.03$) $?$ inconclusive

Late	-	$\equiv$	$\equiv$	$?$	$?$	$?$	$?$	$?$	$?$	$?$	$\neq$
Gated	$\equiv$	-	$\equiv$	$\equiv$	$?$	$?$	$?$	$?$	$?$	$?$	$\neq$
Attention	$\equiv$	$\equiv$	-	$\equiv$	$?$	$\equiv$	$\equiv$	$\equiv$	$?$	$?$	$\neq$
Score	$?$	$\equiv$	$\equiv$	-	$?$	$\equiv$	$\equiv$	$\equiv$	$?$	$?$	$\neq$
Early/Interm.	$?$	$?$	$?$	$?$	-	$?$	$?$	$?$	$?$	$?$	$\neq$
2D-CNN (wide)	$?$	$?$	$\equiv$	$\equiv$	$?$	-	$\equiv$	$\equiv$	$?$	$?$	$\neq$
1D-CNN (wide)	$?$	$?$	$\equiv$	$\equiv$	$?$	$\equiv$	-	$\equiv$	$?$	$?$	$\neq$
2D-CNN	$?$	$?$	$\equiv$	$\equiv$	$?$	$\equiv$	$\equiv$	-	$?$	$\equiv$	$\neq$
1D-CNN	$?$	$?$	$?$	$?$	$?$	$?$	$?$	$?$	-	$\equiv$	$\neq$
Shallow-LR (7 feat.)	$?$	$?$	$?$	$?$	$?$	$?$	$?$	$\equiv$	$\equiv$	-	$\neq$
LogVar (1 feat.)	$\neq$	$\neq$	$\neq$	$\neq$	$\neq$	$\neq$	$\neq$	$\neq$	$\neq$	$\neq$	-
	Late	Gated	Attention	Score	Early/Interm.	2D-CNN (wide)	1D-CNN (wide)	2D-CNN	1D-CNN	Shallow-LR (7 feat.)	LogVar (1 feat.)

Figure 2: Bonn T1, all 11 models, pairwise outcome of the corrected test at $\delta_{\min} \leq 0.03$ macro F1. Green cells (late, gated, attention, score, and the two widened unimodal controls) are established equivalent to each other; orange cells are **logvar** against everything else, the only pairs that reach corrected significance in this dataset. Grey cells, including every comparison involving early fusion, are inconclusive at this margin and sample size, not equivalent and not significantly different. Note that this figure sources the same 2-thread numerical realisation as Table 2 (Section 6.2); the one significant early-vs-late pair reported in Section 4.3 comes from a separate 12-thread realisation and does not appear here, itself a small illustration of the thread count finding.

4.3 Are the fusion operators distinguishable from each other?

Restricting to the five fusion operators (early, late, gated, attention, score) and their ten pairwise comparisons on T1, using the numerical realisation with recorded log loss and Brier (Section 6.2 explains why two realisations exist), the corrected test finds **one significant pair on macro F1** (early vs late, $p_{\text{Holm}} = 0.048$) and **three on Brier** (early vs gated, early vs late, early vs score, all $p_{\text{Holm}} < 0.04$); log loss finds none, consistent with its instability (Section 3.5). Every other pair among late, gated, attention, and score is statistically indistinguishable under the corrected test, with equivalence margins $\delta_{\min}$ between 0.022 and 0.028 macro F1, and Bayesian posterior probability of equivalence within a ± 0.01 ROPE between 0.48 and 0.61 for these pairs, against 0.001 to 0.005 for pairs involving early. In short: **late, gated, attention, and score are equivalent to each other within about 0.02 to 0.03 macro F1; early is not equivalent to any of them, and is worse.**

4.4 Robustness of the null result

Repeating the T1 comparison at three spectrogram resolutions and three fusion parameter budgets (`results_v2/phase0/sens_*`) leaves the conclusion intact at five of six design points (9 of 45 pairs significant at each, all `logvar` related as in Section 4.2); the one partial exception is that score level fusion, the one operator that does not train its two branches jointly, degrades when one branch is deliberately weakened, which is the expected behaviour of an operator that cannot compensate for an underperforming branch during training.

The amplitude normalisation ablation (N1: spectrogram computed from the unnormalised signal, against N2: both branches normalised) shows early fusion shifts by 0.062 macro F1 between the two arms (naive $p < 1e-6$; corrected $p_{\text{Holm}} = 0.067$, borderline after correction) while every other model is statistically untouched (`results_v2/phase0/ablation_N1_vs_N2.csv`). This localises an amplitude related shortcut to exactly the one architecture that concatenates the raw signal into the spectrogram’s channel dimension (Section 3.1), and is a plausible mechanistic explanation for why early fusion is the one operator that separates.

5. CHB-MIT: External Validation

Bonn publishes no per segment subject identifier, so no claim in Section 4 bears on whether a model generalises across patients rather than across segments drawn from a small, possibly overlapping pool. CHB-MIT (Shoeb 2009; Goldberger et al. 2000), publicly available via PhysioNet, provides 24 pediatric recording cases with subject identifiers, making leave one subject out (LOSO) evaluation possible for the first time in this comparison. Two of the 24 cases are documented as the same person, so the evaluation has 23 folds rather than 24 (Section 5.2).

5.1 Data preparation and two errors found in the source data

We downloaded and verified the full corpus (686 recordings, 982.9 hours, `src/verify_chbmit.py`), confirming file integrity against sizes computed from each EDF header rather than trusting file listings.

Annotation labelling error. Summing seizure counts from each subject’s summary file gives 198 seizures, matching PhysioNet’s published count, but the naive approach of pairing “Seizure N Start/End Time” lines by their printed number produces a total ictal duration of 17,856 seconds. One file, `chb09_08.edf`, contains a labelling error in the source data: its second seizure’s end line is itself printed as “Seizure 1 End Time” rather than “Seizure 2”, so a number based parser pairs

the first seizure’s start (2951 s) with the second seizure’s end (9267 s), fabricating a 6316 second seizure that does not exist. We replaced number based pairing with sequence based pairing (each “Start Time” opens a seizure, each “End Time” closes the most recently opened one, and the parsed count is cross checked against each file’s declared “Number of Seizures”) and this corrects total ictal duration to 11,611 seconds, a 35 percent reduction, with every subject’s parsed count now matching its declaration (`src/chbmit.py: parse_summary`).

Index file error. PhysioNet’s RECORDS-WITH-SEIZURES index lists `chb07/chb07_18.edf` as containing a seizure, but that file’s own summary declares zero seizures and no `.seizures` annotation file exists for it on the server (HTTP 404); `chb07_19.edf` has both a declared seizure and a server side annotation file. Both independent sources agree the index file names the wrong recording for this one entry.

Channel montage. Channel sets vary not only between subjects but between recordings of the same subject: part of subject 12’s recordings use a different reference montage (channels such as C3-CS2) that shares no channel with the standard bipolar montage used elsewhere, so a naive intersection across all recordings returns zero channels. We fix the standard 18 channel bipolar montage as a target and drop recordings that do not contain it, retaining 683 of 686 recordings and 185 of 198 seizures; the three dropped recordings and 13 dropped seizures are entirely from the affected part of subject 12’s data.

5.2 Windowing, class balance, and leakage control

Ten second, non overlapping windows. Non overlapping windows are an explicit choice: overlapping, sub window strided windowing (which we considered to increase the positive count) risks placing near identical copies of the same seizure on both sides of a train or test split, an error documented in this literature (Ali et al. 2024). A window is labelled ictal if it overlaps an annotated seizure interval at all, so a window that only partly covers a seizure’s onset or offset is labelled ictal. Of the 1,276 ictal windows, 948 lie entirely within a seizure and 328 (25.7 percent) are such boundary windows, containing a median of 5.0 seconds of annotated seizure (minimum 1.0 second). The alternative, excluding boundary windows, is implemented (`src/chbmit.py: plan_windows, guard_s`) but has not been run, and is listed among the CHB-MIT sensitivity analyses still outstanding (Section 7). Because ictal windows are 0.34 percent of all windows before subsampling, non ictal windows are subsampled per subject to a 4:1 ratio against that subject’s own ictal count, and both this ratio and each subject’s true unsampled recording duration are retained so that clinically meaningful rates (false alarms per hour) can in principle be computed on the true time base rather than the subsampled one (a computation we have not yet performed; Section 7).

Two leakage rules are enforced and checked: no subject’s windows may appear in both train and test, and no single seizure’s windows may be split across train and test. We validated the leakage checker itself by constructing a deliberately leaky split (windows shuffled at random, ignoring subject and seizure identity) and confirming it is flagged on both rules; the real LOSO and grouped k-fold splits used throughout are confirmed leakage free by the same checker (`src/chbmit_corpus.py`), under the person level grouping described next.

Two cases, one person. PhysioNet’s own dataset documentation states that “case chb21 was obtained 1.5 years after case chb01, from the same female subject.” We therefore group subjects by person rather than by case folder: `chb01` and `chb21` form one subject, held out together, which gives 23 LOSO folds over 24 cases (`src/chbmit_corpus.py: SAME_SUBJECT`). The leakage checker validates the grouping directly: applied to the earlier, case based split, it flags exactly the two folds in which one of the pair was tested while the other was in training. One residual uncertainty remains: case `chb24` was added to the corpus later and is not listed in its `SUBJECT-INFO` file, so we cannot confirm from the documentation that it is a distinct person;

we treat it as one. Within each training fold, four of the 22 training subjects are held out, as whole subjects, for the inner validation split used for early stopping and for the score fusion weight.

The resulting corpus is 6,380 windows (1,276 ictal) from 24 cases, 23 subjects, and 18 channels. Model architectures are unchanged from Section 3.1 except that the first convolutional layer of each backbone accepts 18 input channels rather than one; this adds channels only to the input layer and leaves every reported Bonn parameter count identical when re run at 1 input channel, which we verified directly rather than assuming.

5.3 Results

Table 3: CHB-MIT, macro F1, log loss, and Brier, mean over 23 leave one subject out folds.
Source: `results_v2/chbmit/loso_grouped/perfold.csv`.

Model	F1	F1 SD	Log loss	Brier
spec2d_wide	0.673	0.132	0.752	0.291
shallow	0.671	0.180	0.634	0.373
spec2d	0.671	0.121	1.012	0.330
attention	0.668	0.164	0.982	0.318
raw1d	0.668	0.152	0.700	0.311
score	0.666	0.154	0.499	0.274
early	0.666	0.164	1.047	0.354
gated	0.659	0.165	0.965	0.344
raw1d_wide	0.657	0.130	0.806	0.329
late	0.649	0.150	0.804	0.318
logvar	0.622	0.162	0.615	0.416

No pair among the five fusion operators separates under the corrected test on any metric (0 of 10 pairs on macro F1, log loss, or Brier; `results_v2/chbmit/phase0_loso_grouped/chbmit_*.csv`), and in fact **no pair among all 55 comparisons across all 11 models separates on macro F1, even under the uncorrected naive test** (0 of 55). This is a stronger non separation than Bonn’s, where the naive test found 22 of 55 pairs “significant” before correction. The reason is visible in the standard deviation column of Table 3: between subject variability (SD 0.12 to 0.18 in macro F1) is large relative to the mean differences between models (spread across all 11 models of 0.051, and across the five fusion operators of 0.020), so no resampling based test, corrected or not, has the power to resolve differences at this scale from 23 subjects. Across all 55 pairs and all three metrics, the corrected test finds a single difference: score fusion’s lower log loss than raw1d_wide ($p_{\text{Holm}} = 0.048$), which does not survive the conservative test to train ratio ($p_{\text{Holm}} = 0.079$; Section 7) and which we therefore treat as fragile.

This is not the same finding as Bonn’s, and we report it separately rather than folding both into one sentence about operators being “indistinguishable”. On Bonn, equivalence is positively established: the Bayesian posterior probability that late, gated, attention, and score are practically equivalent (within a ± 0.01 macro F1 ROPE) reaches as high as 0.61 for some pairs. On CHB-MIT, the same posterior probability for the same five operators’ ten pairs ranges only 0.19 to 0.31, and the corrected equivalence margin $\delta_{\min}$ ranges 0.049 to 0.079, wider than any Bonn pair. **CHB-MIT does not show that the operators are equivalent; it shows only that this design, at 23 subjects, cannot resolve whether they are equivalent or different.** This is a materially weaker and more honest claim, and the distinction matters for anyone citing this result: “we could not tell the difference” is not “there is no difference.”

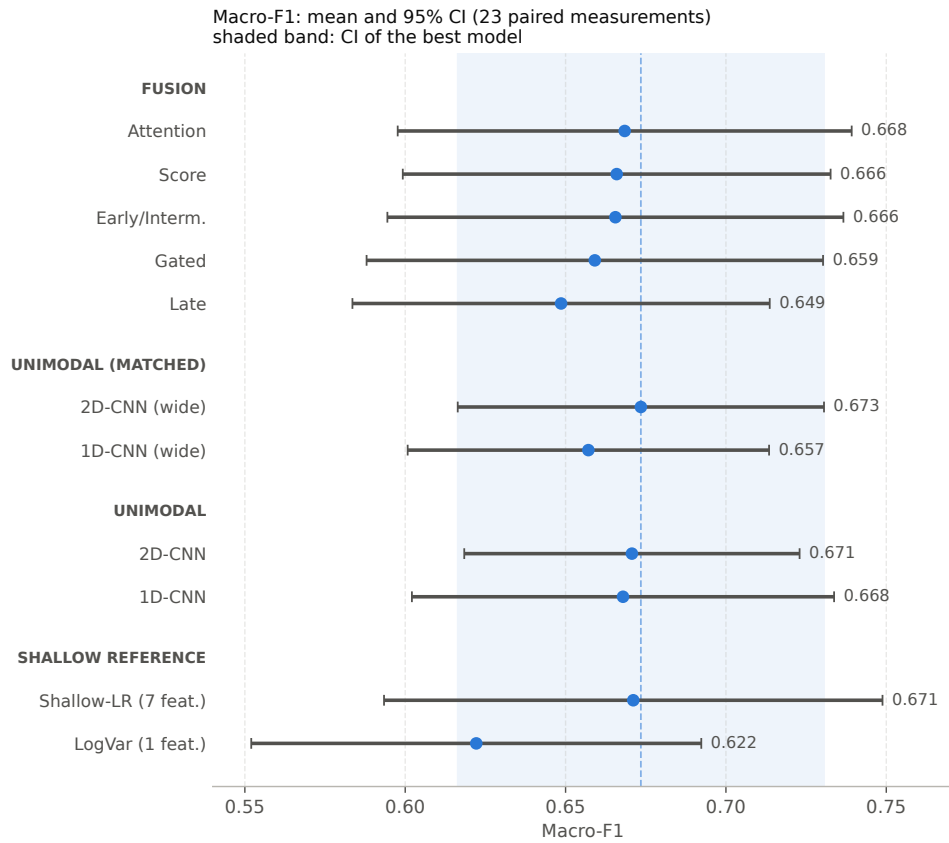


Figure 3: CHB-MIT LOSO, mean macro F1 with 95% CI per model, same axis conventions as Figure 1. Compare the width of these intervals to Figure 1's: every model's 95% CI here spans roughly 0.10 to 0.16 macro F1, wide enough that all eleven overlap substantially, which is the between subject variability problem stated in prose below made visible.

Pairwise comparison: macro F1 (CHB-MIT LOSO) (corrected test)
 $\neq$ significant difference (corrected+Holm, $p < 0.05$) $\equiv$ equivalent ($\delta_{\min} \leq 0.03$) ? inconclusive

Late	-	?	?	?	?	?	?	?	?	?	?
Gated	?	-	?	?	?	?	?	?	?	?	?
Attention	?	?	-	?	?	?	?	?	?	?	?
Score	?	?	?	-	?	?	?	?	$\equiv$	?	?
Early/Intern.	?	?	?	?	-	?	?	?	?	?	?
2D-CNN (wide)	?	?	?	?	?	-	?	?	?	?	?
1D-CNN (wide)	?	?	?	?	?	?	-	?	?	?	?
2D-CNN	?	?	?	?	?	?	?	-	?	?	?
1D-CNN	?	?	?	$\equiv$	?	?	?	?	-	?	?
Shallow-LR (7 feat.)	?	?	?	?	?	?	?	?	?	-	?
LogVar (1 feat.)	?	?	?	?	?	?	?	?	?	?	-

Late Gated Attention Score Early/Intern. 2D-CNN (wide) 1D-CNN (wide) 2D-CNN 1D-CNN Shallow-LR (7 feat.) LogVar (1 feat.)

Figure 4: CHB-MIT LOSO, all 11 models, the same corrected test and the same $\delta_{\min} \leq 0.03$ margin as Figure 2. Every cell but one is inconclusive. The exception, raw1d and score fusion established as equivalent ($\delta_{\min} = 0.026$), is expected rather than informative: score fusion is a weighted average of raw1d’s and spec2d’s predicted probabilities in which raw1d receives a mean validation selected weight of 0.65, so the two make largely the same decisions. Contrasted directly with Figure 2, which is mostly green (equivalent) with a clear orange block (**logvar** significantly worse), this is the clearest single illustration in this paper of the difference between “no difference found” and “equivalence established”: identical method, identical margin, and CHB-MIT’s between subject variance is enough to erase every distinction Bonn’s design could draw, including the one, **logvar** being worse, that Bonn establishes most strongly of all.

A further consequence is that **early fusion’s Bonn specific weakness does not replicate on CHB-MIT**: it ranks seventh of eleven models by macro F1, not last, and the lowest ranked deep model is instead late fusion, the operator with the highest mean on Bonn T1, although no CHB-MIT difference is significant and Section 5.4 shows that this ordering is not stable across runs. We cannot distinguish, with the present data, whether early fusion’s Bonn result reflects something genuine about that architecture or reflects a Bonn specific artefact (Section 4.4’s amplitude shortcut finding is itself Bonn specific, since it concerns the interaction between early fusion and a spectrogram caching bug in the Bonn pipeline the current implementation does not share).

Classical features remain competitive. The **shallow** classical baseline (a logistic regression on seven hand engineered features per channel) ranks second of eleven models by macro F1, 0.002 behind the best model, with a higher mean than all five fusion operators and the `raw1d_wide` control, though none of these differences is significant. It ranked third in the earlier run as well (Section 5.4). This is a less extreme version of Bonn’s saturation finding (Section 4.1) but the same direction: the representation learning component of these models is not obviously earning its keep over a linear classifier on hand engineered features, on either corpus.

Score level fusion is best calibrated, not necessarily most accurate. Score fusion, the simplest of the five operators, an ensemble of two independently trained unimodal models with no additional training, achieves the lowest log loss of all eleven models (0.499, next best 0.615) and the lowest Brier score (0.274, next best 0.291), the latter not significantly different from any other model. Its macro F1 (0.666) is unremarkable and not separable from the other operators. This is also one of the few CHB-MIT patterns that held across both runs (Section 5.4: log loss 0.501 and joint lowest Brier in the earlier run). We flag it as an interesting but narrow finding: classification accuracy and probabilistic calibration are different properties, and an operator that wins on one need not win on the other. This has a plausible theoretical explanation rather than being an accident of this dataset: Mattei and Garreau (2025) show that averaging predictions from an ensemble of models is guaranteed to help a convex loss, by Jensen’s inequality, but carries no such guarantee for a non convex one. Log loss and Brier score are convex, macro F1 is not, which is consistent with score fusion (the one operator here that is a literal post hoc average of two independent models) improving on the two convex metrics without improving on the non convex one.

5.4 Re-running CHB-MIT: what changed and what did not

The results above come from a full re-run. An earlier CHB-MIT run (`results_v2/chbmit/loso_main/`), on which a previous draft of this manuscript was based, differed in four ways: it grouped subjects by case folder, giving 24 folds and letting `chb01` and `chb21` appear on opposite sides of a split; it used Adam rather than AdamW; it ran on a different machine with mixed thread counts (12 threads for the first five folds, 6 for the rest); and it computed score fusion in a separate script. The corrected run changes all four at once, so the differences below cannot be attributed to any single one of them.

The conclusions did not change. In both runs, 0 of 55 pairs separate on macro F1 under either the naive or the corrected test, no pair of fusion operators separates on any metric, equivalence is not established for any fusion pair, and score fusion has the lowest log loss.

The ranking did. Mean macro F1 per model moved by between -0.030 and $+0.027$, and the rank correlation of the eleven models between the two runs is weak and not significant (Spearman $\rho = 0.45$, $p = 0.17$). The best fusion operator was gated in the earlier run (second of eleven) and is attention in this one; gated fell to eighth and late fusion from fifth to tenth.

The cause lies in training, not in the data. On the 22 subjects whose test folds are identical in the two runs (every subject except `chb01` and `chb21`), the per fold macro F1 of the nine deep

models changed by a median of 0.056 and by up to 0.364 (gated fusion on chb05), while the two classical baselines, which involve no neural network training, changed by at most 0.004. The windows, labels, and features are therefore the same; what changed is how the networks trained.

The grouping error distorted two folds, in opposite directions. In the earlier run, the fold testing chb01 (with chb21 in training) was among the best of all folds (late fusion 0.981, its best fold), while the fold testing chb21 (with chb01 in training) was the second worst averaged over the deep models: six of the nine detected no ictal window at all (late fusion 0.444). Held out together, the same person now scores 0.841 with late fusion. The leak therefore did not simply inflate results, and its net effect on the mean was small, but it made two of 24 folds unrepresentative.

We read this as the thread count finding of Section 6.2 at a larger scale: a set of changes with no bearing on which fusion operator is better moves individual models by as much as the entire spread between them. A single CHB-MIT run supports the null result, which is about the absence of separable differences, but it does not support any ranking of the operators, and neither run’s ordering should be cited as one.

5.5 Third corpus: Siena Scalp EEG (in progress)

The Siena Scalp EEG Database (Detti et al. 2020; PhysioNet) holds 41 recordings of 14 adult patients (47 seizures, 512 Hz, referential 10-20 montage; 141.0 hours by the EDF headers). The database description gives about 128 hours, the total of its `subject_info.csv`, which differs from the recordings for four patients: for PN14 it inherits a three-hour error in the seizure list, and for PN03, PN10 and PN12 it is 11.7, 2.0 and 2.0 hours shorter than the files, without an explanation in the data (the files are complete and contain no filler; `docs/SIENA.md`, Section 8). After windowing and the same 1:4 subsampling as CHB-MIT, the corpus has 1,640 windows (328 ictal) and 14 leave one subject out folds, confirmed free of leakage by the same checker. It is converted to exactly the CHB-MIT form, so every driver is shared: the 18 channels of the CHB-MIT bipolar montage are derived from the referential electrodes, the signal is resampled to 256 Hz on the continuous recording, and windowing, labelling and subsampling are CHB-MIT’s own (`src/siena.py`, `src/siena_corpus.py`). Siena serves two purposes: an external replication of the multiverse on adult patients, and, separately from the multiverse, a cross dataset experiment in which models trained on CHB-MIT are tested on Siena. Differences between the corpora will not be interpreted as an age effect, because seizure types, recording structure and acquisition also differ.

The published seizure lists required checks before use (`docs/SIENA.md`). All 41 headers are internally consistent; the listed recording start agrees with the header in 39 recordings. In PN14-3 the list is three hours off and the header is verifiably right (its start plus the recording’s duration gives the listed end exactly). In PN05-3 they differ by 10 s and neither source can be confirmed; using the header there is a decision rule (the header’s start is used for every recording), not a verified correction, and moves the labels of two windows. One seizure end in PN00 lies 32 minutes after its recording ends; we accepted the evident correction (one hour earlier, a 60 s seizure) only after the signal confirmed it: measured with the same rule as the patient’s other four seizures, the ictal rhythm ends 14 s after the corrected end, within the range of those four (0 to 13 s). For PN10 the electrical onset is used where both onsets are listed; it changes the label of one window.

5.6 Multiverse results (pending)

[Pending: the nine pipelines x three seed sets on CHB-MIT are being run; tables and figures will be produced by `python -m src.multiverse --dataset chbmit`: performance per pipeline,

rank stability heatmap, interaction plot, variance components, robust equivalence, robustness against performance, and prediction stability.]

6. Discussion: What Does “Cannot Distinguish” Mean Here

6.1 The naive test is not a minor technicality

Across both corpora, the gap between naive and corrected significance is the single largest effect size in this paper. On Bonn T2 and T3, naive testing reports 20 and 13 “significant” differences respectively that the corrected test reduces to zero. Our simulation (Section 3.5) shows this is expected: the naive test’s false positive rate under repeated cross validation is 37 to 39 percent against a nominal 5 percent, an error rate large enough that a substantial fraction of “significant” findings reported under this common practice, in this literature and adjacent ones, should be presumed spurious until shown otherwise by a corrected re-analysis. We are not aware of prior work in the EEG seizure detection literature specifically that quantifies this by simulation, though the underlying statistical point (Nadeau and Bengio 2003; Bouckaert and Frank 2004) predates this application by two decades.

6.2 Thread count as an unreported confound

While verifying determinism of the CHB-MIT pipeline we found that CPU thread count alone, set via `torch.set_num_threads` and otherwise invisible in any results table, shifts per fold macro F1 by up to 0.093 on Bonn and 0.208 on CHB-MIT (a single outlier fold; median effect near zero, mean effect 0.05 across the 24 fold by 2 model comparisons we made, on the earlier case grouped configuration), holding code, data, and random seed fixed (`docs/10_OLASILIK_VE_TEKRARLANA_BILIRLIK.md`, `docs/13_CHBMIT_ISTATISTIK.md`). We confirmed this is deterministic at fixed thread count (three repeated runs at identical settings reproduce identical results to the last recorded decimal) and that the effect appears only when thread count changes, which localises the cause to floating point summation order in multi threaded BLAS and convolution routines rather than to any nondeterministic scheduling. The consequence for interpretation is direct: on Bonn, the spread across our five fusion operators (0.078 macro F1, Table 2) is comparable to the spread this single, scientifically meaningless setting can introduce on its own (0.093, measured on a separate numerical realisation of the same comparison); on CHB-MIT the thread effect can exceed the entire spread across all eleven models (0.051, Table 3). The re-run of Section 5.4 shows the same phenomenon from a different direction: changing the optimiser, thread count, and machine together reordered the CHB-MIT models while leaving every conclusion intact. The final CHB-MIT run was therefore made on a single machine at one fixed thread count, and the thread count is recorded with every run. Thread count, batch order and library version are not routinely reported in this literature; our finding suggests they should be, and that result tables from single, unreported hardware configurations warrant more caution than they are usually given.

6.3 Two corpora, two different reasons for a null result

We deliberately avoid summarising Bonn and CHB-MIT with the same sentence. Bonn’s task, particularly its conventional binary split, is saturated (Section 4.1; independently confirmed by Kontras et al. 2026), and within that regime the five fusion operators are shown, positively, to be statistically equivalent to each other at a fairly tight margin. CHB-MIT is not saturated in the same way (macro F1 in the 0.62 to 0.67 range, well below ceiling) but its between subject variability is large enough that with 23 subjects, this design has too little power to establish

either equivalence or difference between the operators. Both are negative results for the practical question “does the fusion operator matter”, but they are negative for different statistical reasons, and a reader who only takes away “no difference was found” has lost the more useful, more falsifiable content of each result.

6.4 Defects found and corrected in this project’s own history

Consistent with the standard we apply to the published literature, we report the defects found during this project’s own development rather than omitting them.

Defect	How it was found	Consequence
Spectrogram cache built from the unnormalised signal	Cached values matched the unnormalised spectrogram to numerical precision	Confounded early fusion’s operator effect with an amplitude shortcut (Section 4.4)
Early stopping froze 6 of 55 runs in a trivial single class solution	F1 near 0.40 co-occurring with AUC near 0.96	Fixed with a minimum epoch floor; affected runs recovered to 0.96 to 1.00
Naive paired testing under repeated cross validation	Simulation (Section 3.5)	Most originally reported “significant” differences did not survive correction
CHB-MIT annotation number based parsing	A 27 minute average seizure duration in one subject was not physiologically plausible	Total ictal duration corrected downward by 35 percent (Section 5.1)
CPU thread count changes results	Direct controlled test after an unusual wall clock slowdown was investigated	Documented as a confound (Section 6.2), not fully eliminated
CHB-MIT subjects grouped by case folder	PhysioNet’s documentation states chb21 is chb01 recorded 1.5 years later	Two folds distorted in opposite directions; corrected by grouping by person and re-running all models (Sections 5.2, 5.4)
CHB-MIT trained with Adam, Bonn with AdamW	Documentation of the training protocol	Optimiser unified to AdamW in the re-run (Section 3.4)

7. Limitations

- **The CHB-MIT results of Section 5.3 are a single training run per model.** Section 5.4 shows that the ranking of models is not stable across runs that differ only in settings unrelated to the fusion operator. The multiverse of Section 3.6 trains every pipeline with three seed sets for this reason; until its results are in, no ranking of operators on CHB-MIT should be cited. Case chb24’s identity as a distinct person is assumed rather than documented (Section 5.2).
- **The artefact removal algorithms are not compared on equal terms in one respect.** Infomax and AMICA share one automatic component rule, but GEDAI selects what to remove by its own criterion (Section 3.6), so differences involving GEDAI combine the algorithm with its selection rule. Ocular components are identified without an EOG channel, from frontal proxies, which may also capture frontal ictal activity.
- **The corrected resampled t test assumes a fixed test to train ratio, which an inner validation split complicates.** Del Pup et al. (2025) argue this assumption is frequently violated in deep learning pipelines that hold out an internal validation set, as ours does for early stopping (Section 3.4). Our primary analysis uses the standard ratio $1/(k-1)$ (0.25 for Bonn, $1/22$ for CHB-MIT LOSO). As a sensitivity check we recomputed every test with a conservative ratio based on the data actually used for fitting once the validation split is removed (0.3125 for Bonn; $1/18$ for CHB-MIT, where four of the 22 training subjects

are held out for validation). No conclusion on macro F1 changes: the number of Holm significant pairs is identical in all four analyses (10 of 55 on Bonn T1; 0 of 55 on Bonn T2, Bonn T3, and CHB-MIT), and the corrected equivalence bounds for the fusion operator pairs widen by at most 0.005 macro F1 (`src/phase0.py`, `src/chbmit_stats.py`, column `p_corrected_conservative`). The one CHB-MIT pair significant under the primary ratio, on log loss (Section 5.3), is not significant under the conservative one. This addresses the objection empirically for our design; it does not replace a correction derived for nested splits.

- **The systematic literature search is incomplete.** Its title and abstract screening was done by an AI assistant, checked by a second, independent AI pass (agreement 91.9 percent, kappa 0.81) but not yet by a human reviewer, which a PRISMA conforming review would require; and 122 of the 128 candidates have not been read in full. The gap claimed in Section 2 rests on the closest studies, which were read in full (`docs/15_LITERATUR_SONUCLARI.md`).
- **CHB-MIT’s clinically relevant false alarm rate has not been computed on the true, unsampled time base;** the subsampled rate we could report would not be clinically meaningful and we have chosen not to report it rather than report a number likely to be misread (Section 5.2).
- **No sensitivity analysis has been run on CHB-MIT** corresponding to Section 4.4’s Bonn analysis (window length, frequency ceiling, line noise removal, subsampling ratio, and whether the 25.7 percent of ictal windows that only partly overlap a seizure are kept; Section 5.2); the CHB-MIT results in Section 5.3 should be read as a single design point.
- **Hardware is CPU only**, which bounds both the scale of models we could evaluate and, per Section 6.2, appears to bound the numerical reproducibility of the results themselves.
- **The current implementation is treated by its own authors as preliminary.** The proposal this project is attached to states an intention to rebuild the pipeline cleanly under advisor supervision before treating any of these numbers as final; this manuscript should be read as reporting what that preliminary implementation currently shows, not as a final account.

8. Use of AI Assistance

An AI assistant (Claude) was used substantially in this work: implementation of the data pipeline, model code, and statistical framework; execution of the experiments reported here; drafting of this manuscript; and verification of the reference list against Crossref. The choice of topic, dataset, and research question, the original implementation that was later audited, and the decision at each branch point in the analysis (including which design choices to treat as ablations, and which findings to report as limitations rather than omit) were made by the author, who takes responsibility for understanding and being able to defend every part of this work.

9. Data and Code Availability

All code, the per fold results underlying every number in this manuscript, and the analysis and figure generation scripts are publicly available at github.com/Atytmr07/eeg-fusion-benchmark. `docs/PIPELINE.md` in that repository documents the full pipeline and the command that reproduces each table and figure; the statistics and figures regenerate identically from the stored results with the pinned package versions. Raw EEG data (Bonn, CHB-MIT, Siena) is not redistributed; all three corpora are publicly available from their original sources (Andrzejak et al. 2001; PhysioNet), and the repository documents how to obtain and verify them (`src/chbmit.py`, `src/verify_chbmit.py`, `docs/11_CHBMIT_PLANI.md`, `src/siena_download.py`,

`src/verify_siena.py`, `docs/SIENA.md`). The packages used only to build the P6 caches are listed in `requirements-ica.txt`.

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